

Device	Estimated Average Traffic [Mbps]
Workstation	25
IP phone	0,1
Access Point	300

Concurrency factor 0,7

Floor	Room Type	Number of rooms	Total Estimated Traffic [Gbps]	Estimated Traffic/Room [Gbps]
GROUND FLOOR	A	1	0,38507	385,07
	B	1	0,38507	385,07
	C	5	1,75035	350,07
	D	1	0,70014	700,14
	E	7	3,18549	455,07
	F	1	1,52257	1522,57
	G	2	2,13514	1067,57
	H	2	1,36514	682,57
	I	2	0,77014	385,07
	J	5	1,75035	350,07
	K	6	2,52042	420,07
	L	0	0	420,07
FIRST FLOOR	A	1	0,38507	385,07
	B	1	0,38507	385,07
	C	5	1,75035	350,07
	D	1	0,70014	700,14
	E	7	3,18549	455,07
	F	1	1,52257	1522,57
	G	2	2,13514	1067,57
	H	0	0	682,57
	I	2	0,77014	385,07
	J	5	1,75035	350,07
	K	6	2,52042	420,07
	L	1	0,42007	420,07

Estimated peak traffic value in network [Gbps] 32

DF	Number of WS	Number of AP	Number of IP Phones	
MDF	246	18	16	A
IDF1	227	21	18	1
IDF2	246	18	16	0
IDF3	209	18	17	1
				0

Total outlets 1070

Equipment					
Type	Choice 1 (CISCO)				
	Name	Required number/DF	Price	Link	Name
Wireless Mangement	CW9800L	1	3.686,60 EUR	link	HPE Aruba Networking Central
CHASSIS	Cisco Catalyst 9410R	1	6.100,82 EUR	link	HPE Aruba Networking CX 6410 v2 Switch

SUPERVISOR	C9400-SUP-1XL	1	7.123,14 EUR	link	HPE Aruba Networking CX 6400 Management Module
LINE CARD (WS)	C9400-LC-48P	6	4.959,83 EUR	link	HPE Aruba Networking CX 6400 R0X39C
LINE CARD (IPPHONE)					
OF CONNECTOR	QSFP+	14	343,64 EUR	link	HP JL310A
LINE CARD (OF)	C9400-LC-12QC (12x40GE)	1	17.336,86 EUR	link	HPE Aruba Networking CX 6400 R0X45C
LINE CARD (AP)	C9400-LC-48UX	1	5.015,78 EUR	link	HPE Aruba Networking CX 6400 24p S1T83A
ACCESS POINT	Cisco Wireless 9178	18	5.455,89 EUR	link	HPE Aruba Networking AP-765
FIREWALL	Cisco Secure Firewall 4245	1	152,257.81	link	PALO-ALTO PA-3440
IP PHONES	Cisco IP Phone 8861	16	366,20 EUR	link	HP 82M93AA
CHASSIS	Cisco Catalyst 9410R	1	6.100,82 EUR	link	HPE Aruba Networking CX 6410 v2 Switch
SUPERVISOR	C9400-SUP-1XL	1	7.123,14 EUR	link	HPE Aruba Networking CX 6400 Management Module
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LINE CARD (AP)	C9400-LC-48UX	1	5.015,78 EUR	link	HPE Aruba Networking CX 6400 24p S1T83A
ACCESS POINT	Cisco Wireless 9179F	21	5.455,89 EUR	link	HPE Aruba Networking AP-765
IP PHONES	Cisco IP Phone 8861	18	366,20 EUR	link	HP 82M93AA
CHASSIS	Cisco Catalyst 9410R	1	6.100,82 EUR	link	HPE Aruba Networking CX 6410 v2 Switch
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ACCESS POINT	Cisco Wireless 9179F	18	5.455,89 EUR	link	HPE Aruba Networking AP-765
IP PHONES	Cisco IP Phone 8861	17	366,20 EUR	link	HP 82M93AA

710.978,24 EUR

Public Ip	Private Ip
203.0.113.0	10.1.0.0

VLAN Name	Network ID	Gateway	Last Usable IP	Broadcast	Mask
Management	10.1.10.0	10.1.10.1	10.1.10.254	10.1.10.255	/24
IP_Phone_Data	10.1.19.0	10.1.19.1	10.1.19.254	10.1.19.255	/24
IP_Phone	10.1.20.0	10.1.20.1	10.1.20.254	10.1.20.255	/24
Wireless	10.1.32.0	10.1.32.1	10.1.35.254	10.1.35.255	/22
Data_Staff	10.1.40.0	10.1.40.1	10.1.40.254	10.1.40.255	/24
Data_Guest	10.1.50.0	10.1.50.1	10.1.50.254	10.1.50.255	/24
HR	10.1.100.0	10.1.100.1	10.1.100.62	10.1.100.63	/26
Finance	10.1.110.0	10.1.110.1	10.1.110.62	10.1.110.63	/26

IT_Ops	10.1.120.0	10.1.120.1	10.1.120.62	10.1.120.63	/26
Legal	10.1.130.0	10.1.130.1	10.1.130.62	10.1.130.63	/26
Sales_North	10.1.140.0	10.1.140.1	10.1.140.62	10.1.140.63	/26
Sales_South	10.1.150.0	10.1.150.1	10.1.150.62	10.1.150.63	/26
Sales_East	10.1.160.0	10.1.160.1	10.1.160.62	10.1.160.63	/26
Sales_West	10.1.170.0	10.1.170.1	10.1.170.62	10.1.170.63	/26
Mktg_Digital	10.1.180.0	10.1.180.1	10.1.180.62	10.1.180.63	/26
Mktg_Brand	10.1.190.0	10.1.190.1	10.1.190.62	10.1.190.63	/26
Frontend_Dev	10.1.200.0	10.1.200.1	10.1.200.62	10.1.200.63	/26
Backend_Dev	10.1.210.0	10.1.210.1	10.1.210.62	10.1.210.63	/26
QA_Test	10.1.220.0	10.1.220.1	10.1.220.62	10.1.220.63	/26
Logistics	10.1.230.0	10.1.230.1	10.1.230.62	10.1.230.63	/26
Purchasing	10.1.240.0	10.1.240.1	10.1.240.62	10.1.240.63	/26
Exec_Suite	10.1.250.0	10.1.250.1	10.1.250.62	10.1.250.63	/26
DevOps_Primary	10.2.0.0	10.2.0.1	10.2.0.62	10.2.0.63	/26
DevOps_Second	10.2.10.0	10.2.10.1	10.2.10.62	10.2.10.63	/26
SRE_Internal	10.2.20.0	10.2.20.1	10.2.20.62	10.2.20.63	/26
Security_Ops	10.2.30.0	10.2.30.1	10.2.30.62	10.2.30.63	/26
Training_Lab	10.2.40.0	10.2.40.1	10.2.40.62	10.2.40.63	/26

Number of workstations	Number of IP phones	Number of access points	Number of simple outlets
10	1	1	12
10	1	1	12
8	1	1	50
16	2	2	20
14	1	1	112
63	1	2	66
37	1	2	80
15	1	2	36
10	1	1	24
8	1	1	50
12	1	1	84
12	1	1	0
10	1	1	12
10	1	1	12
8	1	1	50
16	2	2	20
14	1	1	112
63	1	2	66
37	1	2	80
15	1	2	0
10	1	1	24
8	1	1	50
12	1	1	84
12	1	1	14

Total number of simple outlets 1070

Total number of IP Phones 67

Total number of AP 75

Total number of WS 928

Number of Room Types Covered										
B	C	D	E	F	G	H	I	J	K	L
1	5	1	5	1	1	0	0	0	0	0
0	0	0	2	0	1	2	2	5	6	0
1	5	1	5	1	1	0	0	0	0	0
0	0	0	2	0	1	0	2	5	6	1

Choice 2			DF
Required number/DF	Price	Link	
1	subscription	link	
1	26.334,70 EUR	link	

1	14.125,61 EUR	link	MDF
6	5.350,00 EUR	link	
12	1.295,00 EUR	link	
1	47.650,00 EUR	link	
1	15.545,00 EUR	link	
18	1.475,37 EUR	link	
1	68.179,32 EUR	link	
16	194,99 EUR	link	
1	26.334,70 EUR	link	
1	14.125,61 EUR	link	IDF1
6	5.350,00 EUR	link	
1	47.650,00 EUR	link	
12	1.295,00 EUR	link	
1	15.545,00 EUR	link	
21	1.475,37 EUR	link	
18	194,99 EUR	link	
1	26.334,70 EUR	link	IDF2
1	14.125,61 EUR	link	
6	5.350,00 EUR	link	
1	47.650,00 EUR	link	
12	1.295,00 EUR	link	
1	15.545,00 EUR	link	
18	1.475,37 EUR	link	
16	194,99 EUR	link	IDF3
1	26.334,70 EUR	link	
1	14.125,61 EUR	link	
5	5.350,00 EUR	link	
1	47.650,00 EUR	link	
12	1.295,00 EUR	link	
1	15.545,00 EUR	link	
18	1.475,37 EUR	link	
17	194,99 EUR	link	

791.727,64 EUR

Target Use	Max Capacity
254	254
254	254
254	254
1000	1022
254	254
254	254
48	62
48	62

48	62
48	62
48	62
48	62
48	62
48	62
48	62
48	62
48	62
48	62
48	62
48	62
48	62
48	62
48	62
48	62
48	62
48	62
48	62
48	62
48	62

Total target Use
3134

Max IP capacity
3408